NUTM
NIGERIAN UNIVERSITY OF TECHNOLOGY & MANAGEMENT

CYB 305

Computer Forensics Fundamentals

Week 1 Lecture

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Course Textbook and Study Material

EC-Council
Digital Forensics Essentials (DFE)

Module 01: Computer Forensics Fundamentals

The EC-Council DFE course material is the reading reference for this CYB 305 lecture.

Module coverage

Computer forensics • Cybercrimes
Digital evidence • Forensic readiness
Investigator roles • Legal compliance

How to use the material

Read Module 01 alongside the lecture.
Review key definitions and evidence-handling principles before class discussions.

Module Objectives

01 Define the fundamental concepts of computer forensics

02 Identify common types of cybercrimes and their impact

03 Classify different types of digital evidence and their sources

04 Describe the concept of forensic readiness in business continuity

05 Evaluate legal compliance requirements during forensic investigations

Module Flow

01	Introduction to Computer Forensics
02	Digital Evidence
03	Forensic Readiness
04	Role of a Forensic Investigator
05	Legal Compliance in Computer Forensics

What is Computer Forensics?

Computer forensics involves a set of methodological procedures and techniques for identifying, collecting, preserving, extracting, interpreting, documenting, and presenting digital evidence from computing equipment, ensuring its admissibility in legal and/or administrative proceedings.

The primary goal is to reconstruct events, validate facts, and attribute actions related to a digital incident.

Key Characteristics

- Encompasses a vast array of digital devices and data sources, including computers, mobile devices, networks, and cloud storage.
- Follows a strict, documented methodology to ensure the integrity and chain of custody of all evidence, from acquisition to presentation.
- The entire process is conducted to ensure that the collected evidence is reliable, authentic, and can withstand scrutiny in legal or administrative proceedings.
- Supports investigations into a wide range of incidents, including cybercrimes, insider threats, data breaches, malware incidents, and other digital misconduct.

Objectives of Computer Forensics

01 Identify, collect, and preserve digital evidence systematically

02 Analyze and document evidence for legal admissibility

03 Evaluate the impact of security incidents, data breaches, and cybercrimes

04 Minimize losses by supporting incident response

05 Recover deleted, encrypted, hidden data for forensic analysis

06 Ensure compliance with privacy laws and ethical standards

Need and Applications of Computer Forensics

Need

- To ensure the overall integrity and continued existence of IT systems and network infrastructure within the organizations.
- To extract, process, and interpret the factual evidence such that it proves the attacker's actions in court.
- To efficiently track down perpetrators from different parts of the world.
- To protect the organization's financial resources and valuable time.

Applications

- Prepare for incidents by securing and strengthening the defense mechanism as well as closing the loopholes in security.
- Identify the actions needed for incident response.
- Act against copyright and intellectual property theft/misuse.
- Estimate and minimize damage to resources in a corporate setup.
- Set a security parameter and formulate security norms for ensuring forensic readiness.

Introduction to Cybercrimes

Cybercrime refers to any illegal activity involving a computer, its systems, or its applications, ranging from data theft to system disruption.

Common Cybercrimes	Unauthorized access, financial fraud, identity theft, cyber espionage, ransomware, and online scams
Cybercriminal Tools & Methods	Malware, phishing, exploit kits, botnets, ransomware, and social engineering
Key Targets	Individuals (identity theft, fraud, scams), corporations (data breaches, ransomware), governments (cyber espionage, critical infrastructure attacks), and technology (cloud, IoT, virtual systems)

Classification of Cybercrimes

Internal (Insider) Attack

- Committed by trusted individuals with authorized access.
- May be malicious (intentional) or accidental (negligence, policy violations).
- Targets corporate networks or individual systems.
- Perpetrators include current/former employees, contractors, or ex-partners.
- Example: Insider leaks sensitive data for profit.

External Attack

- Launched by outsiders seeking unauthorized system access.
- Targets organizational data, systems, or cloud assets.
- Uses techniques like phishing, exploits, or malware.
- Perpetrators include cybercriminals, hacktivists, or state-sponsored actors.
- Example: Hacker deploys ransomware via phishing email.

Organizational Impact of Cybercrimes

Data & System Compromise	Breach of confidentiality, integrity, and availability (CIA Triad) Unauthorized access and theft of sensitive data (financial, personal, intellectual property)
Operational & Business Disruptions	Downtime caused by ransomware, DoS attacks, or system intrusions Loss of customer and stakeholder trust, leading to reputational risk
Financial & Legal Consequences	Substantial financial losses (legal fees, remediation, downtime) Regulatory penalties for non-compliance (e.g., GDPR, HIPAA, PCI-DSS)

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Introduction to Digital Evidence

Digital evidence refers to electronically stored or transmitted data that holds evidentiary value in legal or investigative contexts.

Key Characteristics

- Volatile and fragile — can be altered or lost without proper handling
- Requires authentication — must maintain integrity and chain of custody
- Can be direct, corroborative, or circumstantial in investigations

According to Locard's Exchange Principle, "anyone or anything, entering a crime scene takes something of the scene with them, and leaves something of themselves behind when they leave"

Types of Digital Evidence

Volatile Data

- Exists in active memory and is lost when the system is powered off.
- Requires live forensics techniques for collection.
- System Information — system time, logged-on users
- Process & Memory Data — running processes, process-to-port mapping, clipboard contents
- Network Activity — active connections, open ports, routing tables
- Command History — executed shell commands

Non-volatile Data

- Stored on persistent storage devices and remains accessible after power loss.
- Can be recovered using disk forensics techniques.
- File System Artifacts — deleted files, hidden files, unallocated space
- System Logs & Configurations — registry settings, event logs, swap files
- Storage Artifacts — slack space, unused partitions, hidden partitions

Forensic Significance of Digital Evidence

01 Identity theft & financial fraud

02 Malicious attacks on computing systems

03 Information leakage & corporate espionage

04 Unauthorized transmission of confidential data

05 Theft of trade secrets & intellectual property

06 Cyber-enabled criminal activities (e.g., online harassment, fraud)

07 Forgery & falsification of records

08 Unauthorized encryption/password protection of documents

09 Digital evidence in conspiracy (e.g., email communications)

Sources of Digital Evidence

User-Created Files

- Documents: text files, spreadsheets, presentations
- Multimedia: images, audio, video files
- Personal data: address books, databases, internet bookmarks

User-Protected Files

- Encrypted, compressed, or password-protected files
- Hidden or misnamed files, steganographic content

Computer-Created Files

- Logs: system, security, application logs
- Temporary & swap files, backup files
- Configuration settings, browser cookies, registry entries

Sources of Digital Evidence (Cont'd)

Storage Devices	Hard Drive: Text, multimedia, databases, program files Thumb Drive: Text, graphics, images Memory Card: Logs, text files, images, browsing history Removable Media (USB flash drives, external hard drives): Archived data, backups
Computing & Mobile Devices	RAM & Volatile Storage: System processes, network data, active sessions Smartphones/Tablets: Contacts, emails, messages, media, GPS data Handheld Devices (PDAs, Digital Watches): Address books, notes, schedules
Network & Communication Devices	Routers/Switches: Traffic logs, configurations Network Interface Card (NIC): MAC address, connection logs VoIP Systems (e.g., Skype): Call history, messages, contacts, file transfers

Sources of Digital Evidence (Cont'd)

Authentication & Security Devices	Smart Card/Dongle: Access credentials, user authentication data Biometric Scanners: Fingerprint records, access logs Credit Card Skimmers: Cardholder details, transaction records
Multimedia & Peripheral Devices	Digital Cameras: Images, videos, timestamps, metadata Scanners & Copiers: Documents, usage logs Fax Machines: Sent/received logs, stored documents
Emerging Technologies & IoT	IoT & Wearables: GPS, cloud data, sensor logs GPS Devices: Routes, waypoints, travel logs Messaging Apps: Chat history, media, geolocation data

Rules of Evidence

Digital evidence collection must be governed by five basic rules that make it admissible in a court of law:

01	**Admissible**	Evidence must be related to the fact being proved
02	**Authentic**	Evidence must be real and appropriately related to the incident
03	**Reliable**	There must be no doubt about the authenticity or veracity of the evidence
04	**Complete**	The evidence must prove the attacker's actions or his/her innocence
05	**Believable**	The evidence must be clear, credible, and verifiable to ensure jury comprehension and authenticity

Best Evidence Rule

Original Evidence

The Best Evidence Rule states that original evidence (document, photograph, or recording) must be presented in court whenever possible.

Duplicate Evidence

Duplicate evidence may be accepted only if:

- The original is lost or destroyed without bad intent
- It is impractical to produce the original (e.g., large datasets)
- The duplicate is verified as an exact and reliable copy

This rule ensures that the presented evidence is authentic and reliable for legal proceedings.

Federal Rules of Evidence (FRE)

The Federal Rules of Evidence (FRE) govern the admissibility of digital evidence in U.S. courts, ensuring:

FRE Objectives

- Fairness in legal proceedings: Evidence must be relevant and legally obtained
- Elimination of unnecessary delays: Streamlining digital evidence presentation
- Development of evidence law: Adapting legal standards to evolving technologies

Key FRE Provisions

- Rule 401: Evidence must be relevant to the case
- Rule 901: Digital evidence must be authenticated
- Rule 1002 (Best Evidence Rule): The original digital record is preferred
- Rule 1003: Duplicates may be accepted if proven reliable

These rules ensure the credibility, integrity, and admissibility of digital evidence in forensic investigations.

https://www.rulesofevidence.org

Scientific Working Group on Digital Evidence (SWGDE)

Principle 1: To ensure accuracy and reliability in digital evidence handling, forensic agencies must establish and maintain an effective quality management system.

Standards & Criteria for Digital Evidence Handling

Standard 1.1	Agencies must establish and maintain comprehensive SOPs (Standard Operating Procedures) for digital evidence handling.
Standard 1.2	Annual review of SOPs by agency management to ensure continued relevance and effectiveness.
Standard 1.3	Procedures must be widely accepted in the forensic field or scientifically validated.
Standard 1.4	Agencies must maintain written records of technical procedures.
Standard 1.5	Agencies must utilize validated hardware and software for evidence seizure and examination.
Standard 1.6	All activities related to the seizure, storage, examination, and transfer of digital evidence must be documented in writing and available for review and testimony.
Standard 1.7	Any action that may alter, damage, or destroy evidence must be performed by qualified professionals in a forensically sound manner.

https://www.swgde.org

ACPO Principles for Handling Digital Evidence

The Association of Chief Police Officers (ACPO) Principles ensure the integrity, reliability, and admissibility of digital evidence in forensic investigations.

Principle 1 Law enforcement personnel must not alter or modify data stored on computers or digital media that may be used as evidence in court.

Principle 2 If accessing original data is necessary, the person must be competent and able to explain their actions in court.

Principle 3 An audit trail should be created and preserved, allowing an independent third party to replicate the results.

Principle 4 The lead investigator (case officer) holds ultimate responsibility for ensuring compliance with legal standards and ACPO principles.

https://www.npcc.police.uk

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Introduction to Forensic Readiness

Forensic readiness is the proactive capability of an organization to collect, store, and utilize digital evidence efficiently, enabling rapid and cost-effective investigations with minimal operational disruption.

Objectives of Forensic Readiness

01 Enables fast and effective investigations with minimal business disruption

02 Enhances security against intellectual property theft, fraud, and extortion

03 Reduces investigation time and costs through organized evidence storage

04 Facilitates smoother interactions with law enforcement agencies

05 Empowers organizations to use digital evidence in their defense

Forensic Readiness and Business Continuity

Forensic readiness supports business continuity by enabling the swift identification of security incidents, preserving legally admissible digital evidence, and ensuring minimal disruption to critical operations.

Advantages of Forensic Readiness

- Quick identification and classification of cybersecurity incidents
- Efficient evidence collection to support forensic investigations
- Minimizes resource expenditure by streamlining investigation processes
- Enhances damage control and system recovery, reducing downtime
- Strengthens insurance claims with verifiable digital evidence of an attack
- Supports legal action against perpetrators so organizations can claim damages

Consequences of Lacking Readiness

- Extended system downtime, leading to financial losses
- Failure to identify data breaches, resulting in unmitigated security threats
- Reputational damage, leading to client and stakeholder distrust
- Inability to collect admissible digital evidence, weakening legal cases
- Increased vulnerability to data theft, manipulation, and destruction

Forensics Readiness Planning

Forensic readiness planning entails proactive processes to establish and sustain an organization's capacity for effective digital forensic investigations.

01 Define primary goals for forensic readiness

02 Identify potential types of digital evidence for security incidents

03 Determine sources of digital evidence

04 Establish policies for legally admissible evidence extraction

05 Develop secure evidence handling and storage protocols

06 Maintain an incident response team for evidence preservation

07 Secure legal guidance for investigation compliance

08 Implement centralized logging across critical systems for evidence integrity

09 Train personnel on forensic readiness procedures

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Need for a Forensic Investigator

Forensic investigators leverage their expertise in addressing cybercrime, preserving digital evidence, ensuring its admissibility in court, and supporting organizations during security incidents.

Addressing Cybercrime Proliferation

With rising cybercrime rates (e.g., ransomware attacks, fraud), forensic investigators are essential for identifying perpetrators and prosecuting offenses.

Handling Digital Evidence

Most modern investigations rely on electronic evidence such as emails, logs, or network traffic. Forensic investigators ensure its authenticity and integrity.

Investigation Expertise

Leveraging advanced technical skills and experience, forensic investigators assist law enforcement agencies and organizations in uncovering digital traces left by attackers.

Evidence Integrity and Legal Admissibility

Improper handling of digital evidence by untrained personnel can render it inadmissible in court. Forensic investigators use standardized methods to prevent this.

Need for a Forensic Investigator (Cont'd)

Navigating Complex IT Environments

Modern IT environments often involve encrypted data, cloud storage, IoT devices, and large-scale networks. Forensic investigators possess specialized expertise to analyze these systems effectively.

Incident Handling and Response

They play a key role in maintaining forensic readiness by implementing logging mechanisms and preparing incident response plans.

Preservation of Evidence

Forensic investigators ensure evidence is collected without altering its original state using tools like write-blockers and imaging software.

Expert Testimony

Forensic investigators often serve as expert witnesses in court proceedings, explaining complex technical findings to judges and juries.

Key Responsibilities of a Forensic Investigator

01 Determine the scope of data breaches, system intrusions, or unauthorized access

02 Use write-blockers and forensic imaging tools (e.g., FTK Imager) to create bit-for-bit copies of original evidence

03 Recover deleted, hidden, or encrypted data relevant to the investigation

04 Analyze file systems, logs, and metadata to reconstruct incident timelines

05 Provide expert testimony in judicial or administrative proceedings

06 Document evidence handling from collection to courtroom to ensure admissibility

07 Share threat intelligence (e.g., new ransomware variants) with organizations to bolster defenses

08 Advise the organization on incident patterns, forensic readiness, and evidence preservation best practices

What Makes a Good Computer Forensics Investigator?

01 Conduct thorough interviews with clients, victims, witnesses, and suspects to gather critical information about the case

02 Prepare clear, concise reports detailing findings and analysis for legal and organizational stakeholders

03 Use advanced analytical techniques to identify, extract, and link digital evidence to suspects or crimes

04 Effectively explain complex technical findings to non-technical audiences, including judges, juries, and clients

05 Remain current with emerging methodologies and technologies in digital forensics, such as AI-driven analysis and cloud computing

06 Be proficient in multiple operating systems (Windows, macOS, Linux) and familiar with various mobile platforms

07 Have a deep understanding of hardware, software, and networking technologies relevant to digital investigations

08 Develop and maintain professional relationships with experts in computing, networking, and law enforcement to leverage collective expertise

09 Understand relevant laws and regulations governing digital evidence collection and analysis

Common Career Roles in Digital Forensics

Digital forensics offers a wide range of professional opportunities across law enforcement, corporate security, incident response, and specialized technical domains

01 Computer Forensic Investigator

02 Digital Forensic Analyst

03 Incident Response Analyst/Handler

04 Malware Analyst

05 Mobile Device Forensic Expert

06 Law Enforcement Cybercrime Specialist

07 Digital Forensic Technician

08 Cloud Forensic Specialist

09 Forensic Computer Analyst

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Computer Forensics and Legal Compliance

Legal compliance in computer forensics ensures that digital evidence is collected, handled, and analyzed in a lawful and methodical manner, allowing it to be admissible in court and defensible under scrutiny.

Forensic professionals must adhere to applicable laws and industry standards to preserve the integrity and legality of their investigations.

Key Regulations and Standards

1. Gramm-Leach-Bliley Act (GLBA, 1999)
2. Federal Information Security Modernization Act (FISMA, 2014)
3. Health Insurance Portability and Accountability Act (HIPAA, 1996)
4. Payment Card Industry Data Security Standard (PCI DSS)
5. Electronic Communications Privacy Act (ECPA, 1986)
6. General Data Protection Regulation (GDPR, 2018)
7. Data Protection Act (DPA, 2018)
8. Sarbanes-Oxley Act (SOX, 2002)

Other Laws Relevant to Computer Forensics

United States	Foreign Intelligence Surveillance Act	https://bja.ojp.gov
United States	Protect America Act of 2007	https://www.congress.gov
United States	Privacy Act of 1974	https://www.justice.gov
United States	National Information Infrastructure Protection Act of 1996	https://www.congress.gov
United States	Computer Security Act of 1987	https://www.congress.gov
United States	Freedom of Information Act (FOIA)	https://www.foia.gov
United Kingdom	Regulation of Investigatory Powers Act 2000	http://www.legislation.gov.uk
Australia	Cybercrime Act 2001	https://www.legislation.gov.au
Australia	Information Privacy Act 2014	https://www.legislation.gov.au
India	Information Technology Act	http://www.dot.gov.in
Germany	Section 202a. Data Espionage, Section 303a. Alteration of Data, Section 303b. Computer Sabotage	http://www.gesetze-im-internet.de
Italy	Penal Code Article 615 ter	https://www.normattiva.it
Canada	Canadian Criminal Code Section 342.1	https://laws-lois.justice.gc.ca
Singapore	Computer Misuse Act	https://sso.agc.gov.sg
Belgium	Computer Hacking	http://www.ejustice.just.fgov.be
Brazil	Unauthorized modification or alteration of the information system	https://www.planalto.gov.br
Philippines	Data Privacy Act of 2012	https://www.privacy.gov.ph
Hong Kong	Cap. 486 Personal Data (Privacy) Ordinance	http://www.pcpd.org.hk

Module Summary

01 Defined the core principles of computer forensics, emphasizing its role in investigating cybercrimes and securing digital evidence

02 Identified various categories of cybercrimes, such as hacking and fraud, along with their potential impact on organizations

03 Examined the nature of digital evidence, including its sources, types, and the legal principles that govern its admissibility in court

04 Explained the concept of forensic readiness and its significance in ensuring effective incident response and maintaining business continuity

05 Described the key responsibilities and skills of forensic investigators, while exploring relevant career opportunities in the field

06 Reviewed essential legal and regulatory frameworks, such as HIPAA, GDPR, and SOX, that influence computer forensics practices